

Population Composition



RRHS 2019-2020



Population Composition

- Total population tells limited story
- Population composition reveals more
 - How old/young the population is
 - Ratio of men to women
 - Ratio of very old to working ages
 - Ratio of very young to working ages

By now, you know WHERE the big clusters of people are, and this set of notes has a focus on the make-up of these populations.

Why does it matter?

Political practices: Where people live has a large influence on how they vote. In the U.S., not only does the distribution of the population determine how many representatives each area gets, but where people live also is a good predictor of HOW they will vote. Rural areas (less densely populated) tend to vote more conservatively than more urban areas (densely population areas). Politicians in urban areas often campaign on promises of public services, infrastructure development, and other things that appeal to urban voters. Rural areas often don't have these same needs and politicians campaign in a way that appeals to their wants and needs that differ from urban life.

Economic processes: population density is a very big determinant for where businesses will be located. Fewer businesses or jobs are located in rural areas, as there are not enough people in those places to support the businesses financially – you'll learn much more about this pattern in our urban unit next semester.

Social Processes are also influenced by population distribution – The more students a school has, the more diversity and variety of classes the school is able to offer as there will be enough students for these different classes to make. Hospitals in large urban areas are able to have much more advanced medicine and diagnostic equipment available in large urban areas, while rural-area hospitals tend to be only equipped to handle minor-emergencies or routine care, such as broken bones. People in rural areas who experience life-threatening emergencies are often rushed into the nearest large city so they can get the level of care they need, even if it means passing several smaller hospitals along the way.

Knowing the composition of a population--at all scales of analysis, from neighborhoods to world regions—helps us understand each place in context of the people who live there. This class is all about the people, after all.



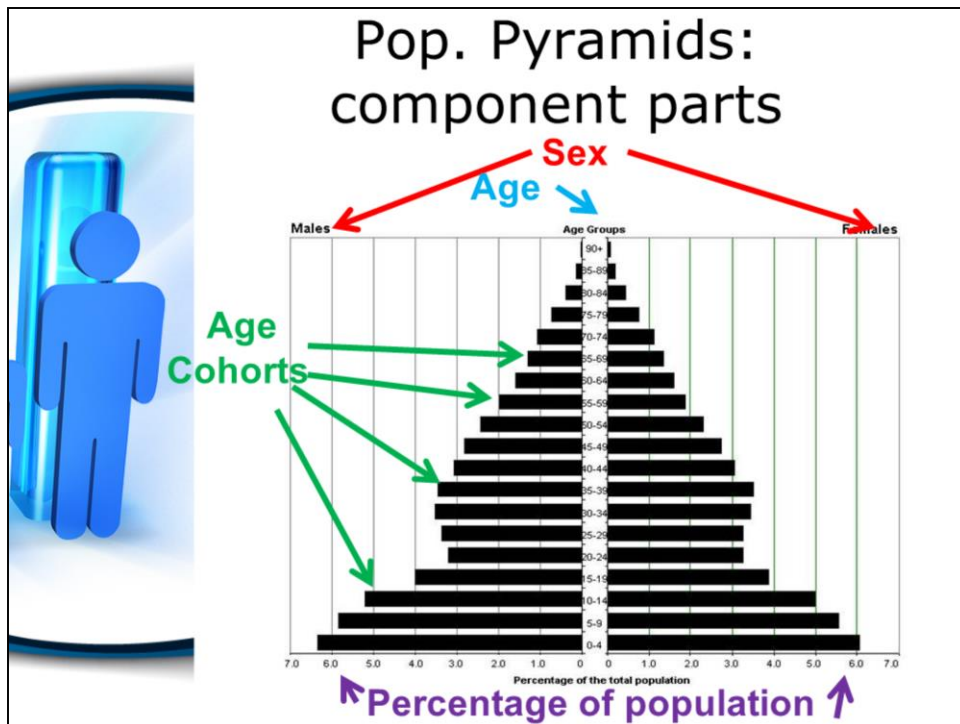
Population Pyramids

- Visual representation of population composition

Population composition can be visualized with a specific kind of graph – the **population pyramid**



No, not this type of pyramid, though if we were to look at the composition of Egypt's population at the time the pyramids were built, we might gain some insight into the social structure of the society.



There are three component parts to every population pyramid: The age cohorts (groups of people within the same age group), the vertical axis (age ranges), and the horizontal axis, which either displays the total number of the population or the percentage of the population.

The sexes are separated in population pyramids – males will be on one side, and females will be on the other. On the pyramid above, the males are on the left, and the females are on the right. Most of the time, these things are in color and will use different colors, but if it's not, check the labels.

- If it is not labeled, look at the very top layers of the pyramid to see which side has more people – that will be the female side. (this is a normal pattern that exists across most societies– women, on average, tend to have a longer life expectancy).

Age may be on the side or the middle but is always the vertical axis.

Age Cohorts are displayed horizontally, stacked from youngest to oldest. Refer to the Age axis to determine the size of the cohort (age group). The most common cohort size is 5 years. This means every person within that cohort will have been born within the same 5 year time period (example: 2001-2005, 2006-2010, etc.)

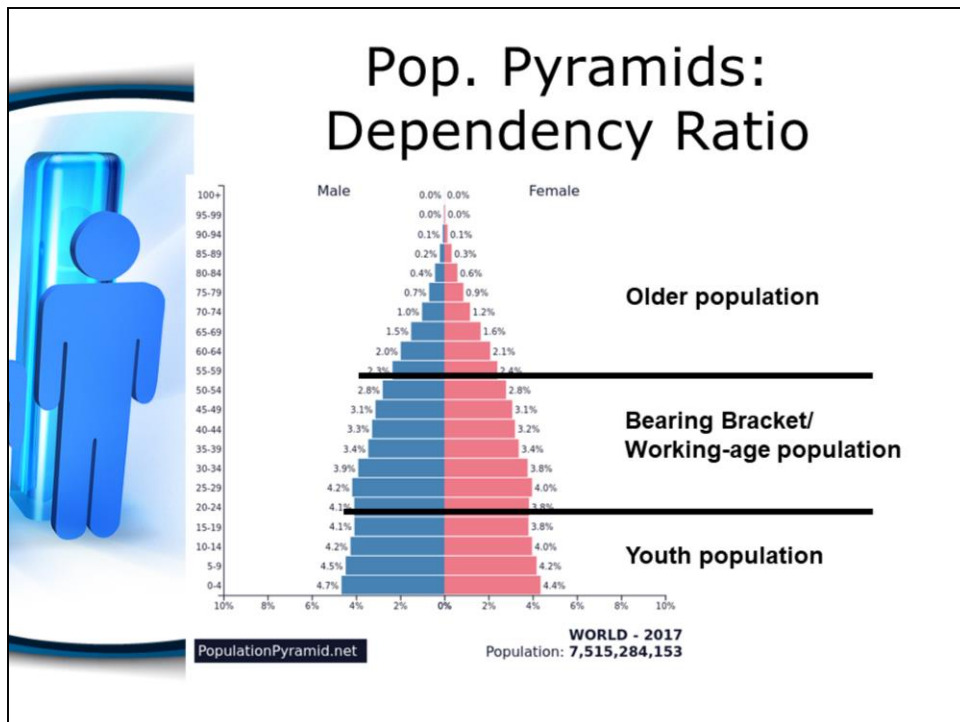
Check the labels to determine how the data is displayed (i.e., by total count, per 1,000, or percentage)



Pop pyramids: Age structure

- Young or old population
- Even or unbalanced?
 - Aging population (more old over 60)
 - Young population (more young under 15)

When we look at the age structure of the population, we're really looking at the ratio between the oldest 1/3 and the working age population (between 15 and 60), or the youngest 1/3 and the working age group. We call this the dependency ratio (more on next slide)



The Dependency Ratio is the balance between the “bearing bracket” – those of working age, 15-60ish, who “bear” the economic, social, and political burden of the society.

The bearing bracket must:

- Care for the children
- Provide the income for their families
- Pay into their employer’s retirement system to provide salaries and healthcare for retirees.
- Pay the vast majority of taxes to the government for government services such as infrastructure, government operating costs, welfare/social security, military expenditures, etc.
- Care for their elderly parents

If there is a significant unbalance – many more children than adults or many more older adults than working age or children, there are going to be social, economic, and political issues associated with that pattern (more discussion on in the notes “Aging populations – coming soon to a website near you).



Pop pyramids: sex ratio

- More males than females?
 - Possible preference for male children
 - Are they guest workers?
 - Is it a military base?
- More females than males?
 - Was there a war?
 - Is it a college town with an all female campus?

When we look at the ratio between the men and women, we're looking for evenness. If the ratio is uneven, we want to know WHY. Geographers get a clue to this WHY when we look at WHERE it is occurring.

Examples: Some cultures, such as China, have a well-known preference for male children. When the Chinese government put restrictions on the number of children families can have, a significant imbalance of male-to-female children resulted.

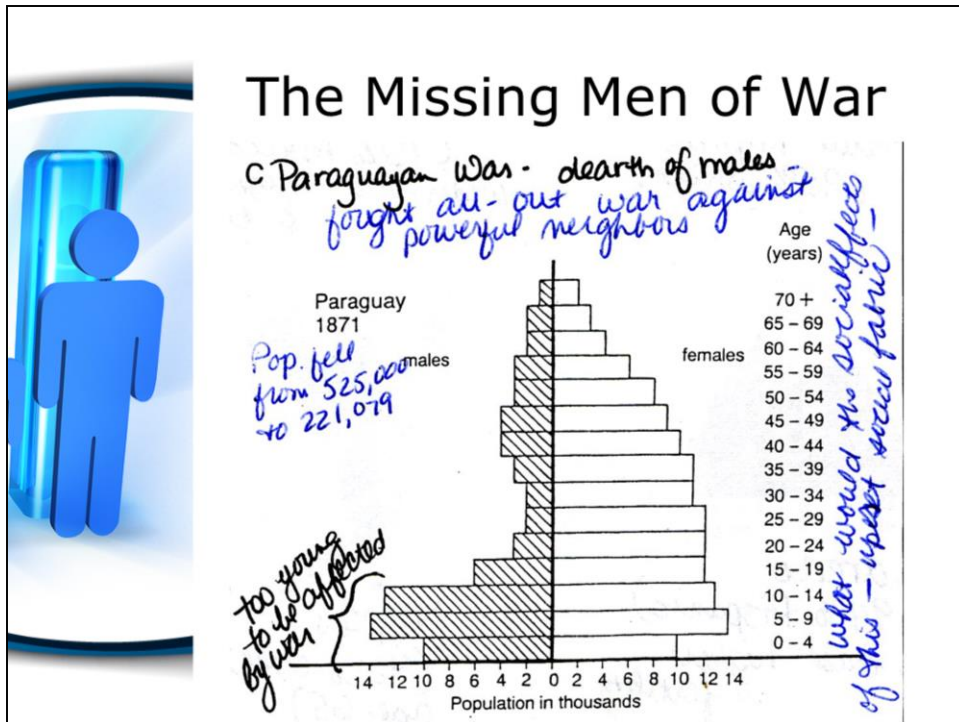
Other places, such as Qatar, an oil-rich nation of Southwest Asia, is experiencing rapid economic growth. They do not have the population of adults to meet the demands of the developers (construction, especially), so they bring in "guest workers" from other countries, primarily places like India. These guest workers are virtually all single males (single women are not permitted to be guest workers in construction). This will cause a significant skew on the population pyramid.

If we zoom in to a city, there may be a large imbalance caused by a military base (mostly males between ages 18-40).

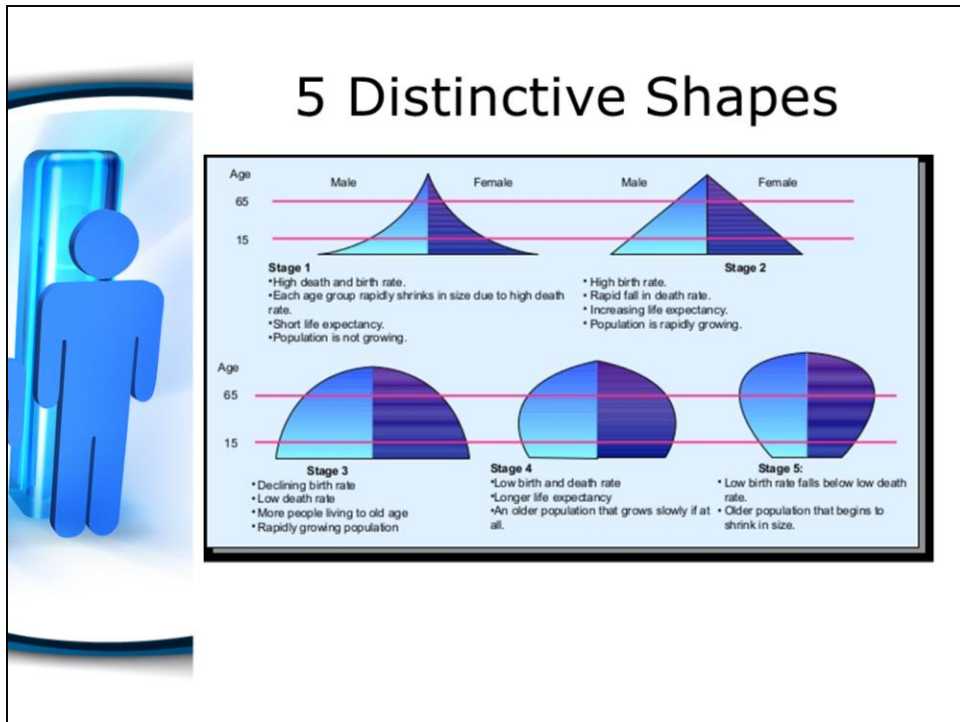
OR, it could be a college town with a disproportionate number of young adults between ages 18-25.

If there is a significant number of more females than men, what happened to the men? If it were an illness, likely there would be a dip on both sides, but if just the men are missing it was likely due to war.

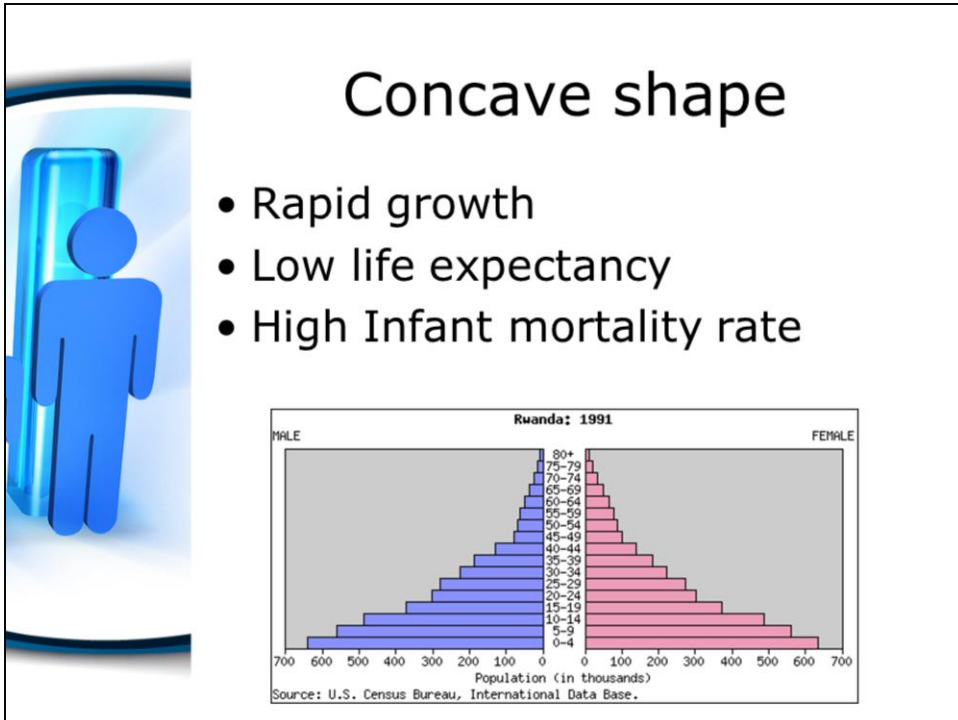
Examples coming up



While this example of Paraguay's population pyramid 1871 is extreme, you can clearly see the pattern.



More about each one coming up...



A pyramid with a wide base shows there are many children being born in each of the youngest age cohorts. The collapsed sides, however, indicate a high mortality rate

Development indicators associated with this pyramid shape:

Literacy Rate: LOW

Life expectancy: LOW

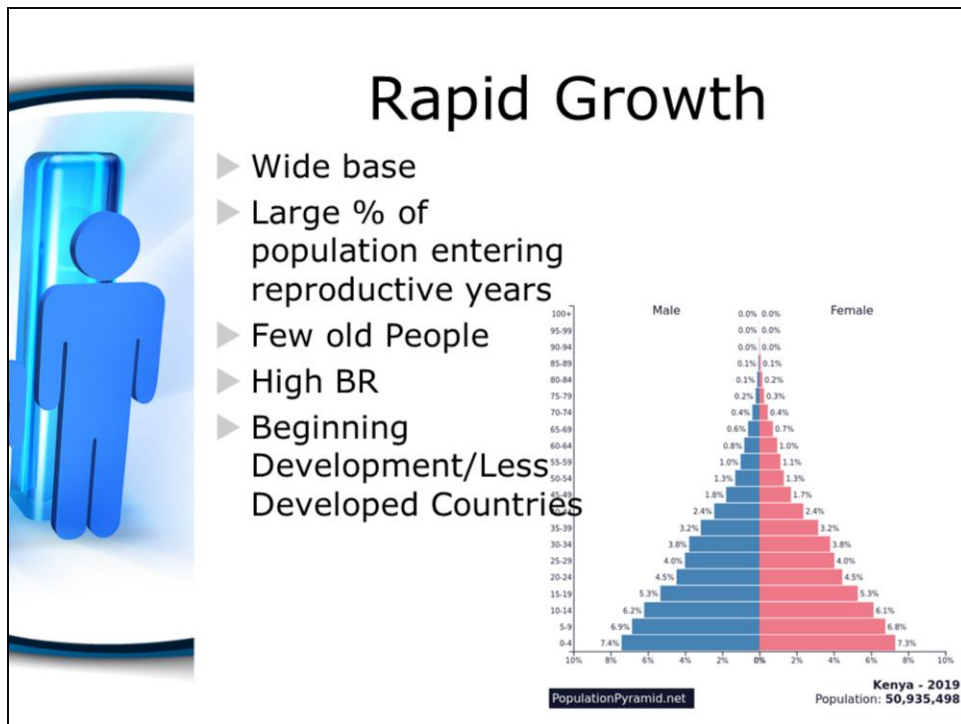
Infant Mortality Rate: HIGH

Per Capita Income: LOW

Birth/Death Rates: HIGH

Total Fertility Rate: HIGH

Education levels for women: LOW



A pyramid with a wide base shows there are many children being born in each of the youngest age cohorts.

As they reach maturity, they will have their own children, likely large families (High Total Fertility and Birth Rate), which will continue to cause the youngest age cohorts to be significantly larger than the older cohorts.

At the same time, improvements for a stable food supply and modern medicine has led to a sharp decline in the Death Rate -- people live longer, fewer are dying from starvation, they are generally healthier so their bodies are better able to fight off diseases -- so a greater percentage of children being born survive childhood to reach adulthood and then reproduce.

This is a rapidly growing population. It will have a high TFR, it will have a high CBR.

Please note that this pyramid shape DOES NOT mean that kids are dying before they reach the next age group. It means that RIGHT NOW (or at the time the "snap shot" was taken), there were that many more kids in the

younger cohorts than in the older ones.

Development indicators associated with this pyramid shape:

Literacy Rate: LOW

Life expectancy: LOW

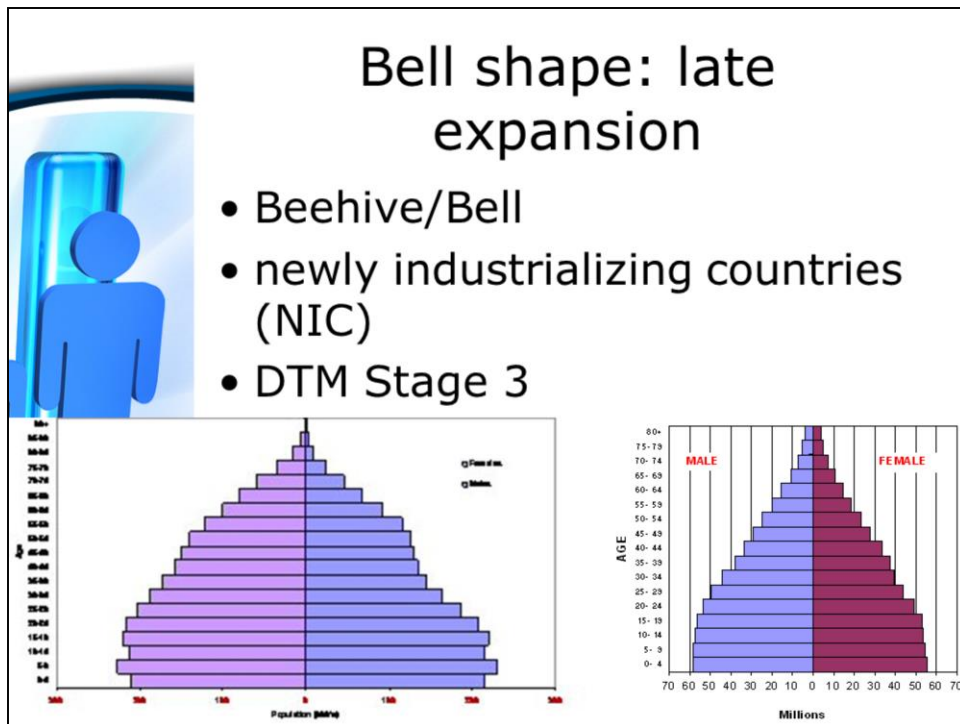
Infant Mortality Rate: HIGH

Per Capita Income: LOW

Birth/Death Rates: HIGH

Total Fertility Rate: HIGH

Education levels for women: LOW



In this stage, growth is still occurring, but is beginning to slow. This is characteristic of Newly Industrialized Countries – the ones who are currently reaching industrial maturity (and shifting from an agricultural based economy to a secondary/tertiary economy) – more people are leaving the rural areas and moving to the cities where more healthcare and other services are available – even if they are not as available or well-developed as they are in MDCs, they are improvement over nothing.

Death Rates are low, and the birth rate is beginning to decline. Women are having fewer children, often by choice.

The youngest age cohorts are about the same size as those right above them....eventually, as the trend continues, this pyramid will morph into a square.

Development indicators associated with this pyramid shape:

Literacy Rate: Moderate, but increasing

Life Expectancy: Increasing

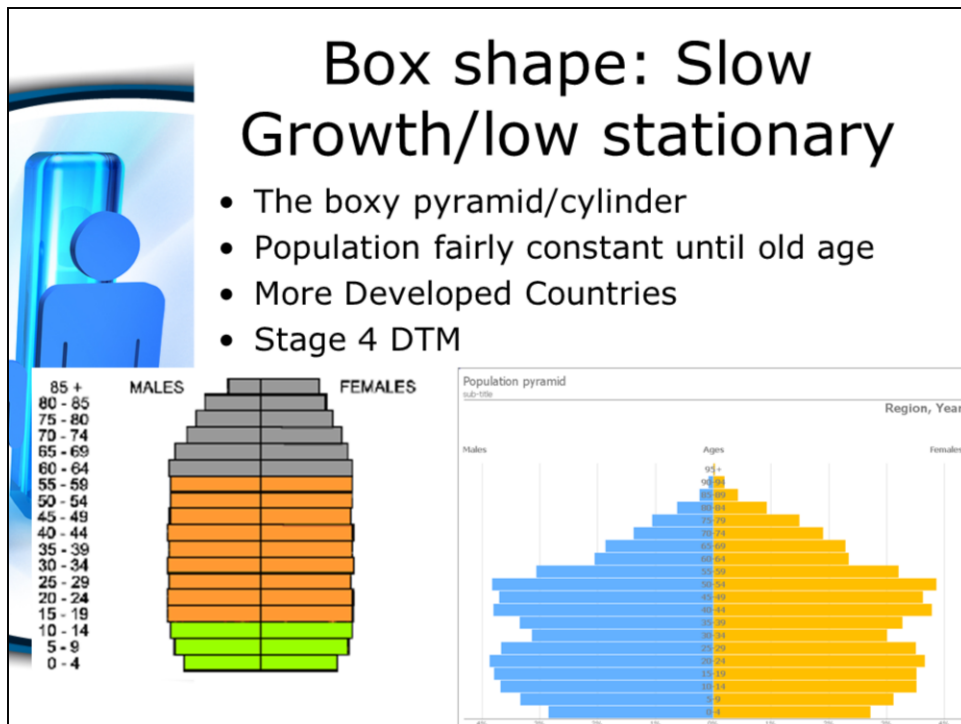
Infant Mortality Rate: Decreasing

Per Capita Income: Increasing

Birth Rate/Death Rate: Falling/low

Fertility Rate: Falling rapidly

Education levels of women: Increasing in urban areas



The square---ish. This means the population size is stable – it is neither growing nor shrinking, and there are about the same number of people across all age cohorts.

Birth rates are low, being at or just above replacement rate (TFR 2.1 children per woman)

These countries are typically urbanized and children are expensive (not a financial asset – they cost money, not make money for the family)

Please note that ALL pyramids with taper off to a triangle at the top because death will eventually come, even if life expectancy is long. To recognize this shape, look at the age cohorts from birth through about age 60 (cover up the top part if you need to).

Development indicators associated with this pyramid shape:

Literacy Rate: High (approaching or at 100%)

Life Expectancy: High (on average 75-80 years)

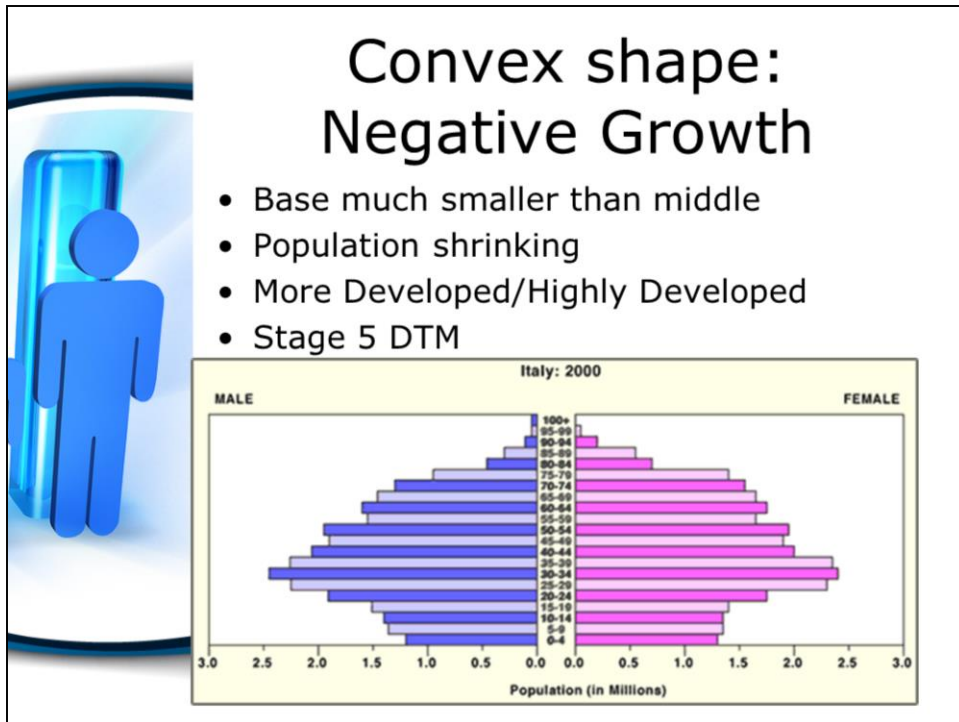
Infant Mortality Rate is LOW (babies are expected to survive infancy)

Per Capita Income: HIGH

Birth Rate/Death Rate: LOW – Birth Rate equal or only slightly higher than death rate

Total Fertility Rate: Low (at replacement rate)

Education Levels of Women: Universal (available to all)



As the age cohorts move up the graph, each successive generation is considerably smaller.

The Natural Increase Rate is **NEGATIVE**, which means the population is getting smaller.

As time progresses and new charts are made to update the population, the pyramids become inverted (tornado or diamond shaped)

Development indicators associated with this pyramid shape:

Literacy Rate: High (approaching or at 100%)

Life Expectancy: High (on average 75-80 years)

Infant Mortality Rate is **LOW** (babies are expected to survive infancy)

Per Capita Income: **HIGH**

Birth Rate/Death Rate: **LOW** – Birth Rate lower than Death Rate in Negative Growth Countries

Total Fertility Rate: Extremely low (at or below 2.1)

Education Levels of Women: Universal (available to all)



How are they used?

- Assess “health” of a population
- Assess predicted population growth
- Predict markets for goods and services

By “health” in this case, geographers and demographers mean stability and growth. Is there a good balance in the dependency ratio?

If the answer is no, the shape of the pyramid gives us a clue –

Youthful populations are often socially and economically underdeveloped – there may be a lack of healthcare, limited access to education, and limited access to sanitation. A large percentage of the economy likely works in the primary sector of the economy (agriculture). Economically, there may not be enough jobs for the kids as they become adults, and there is likely a rapid rural-to-urban migration.

Aged population often have too much burden on the bearing bracket for social services such as healthcare for the elderly (hospitals and nursing homes), as well as economic issues such as not enough people to fill jobs, which can result in economic instability and companies moving their businesses to other countries.



Population Momentum

- How fast is the population growing?
- Period after TFR drops to replacement rate until zero population growth is achieved

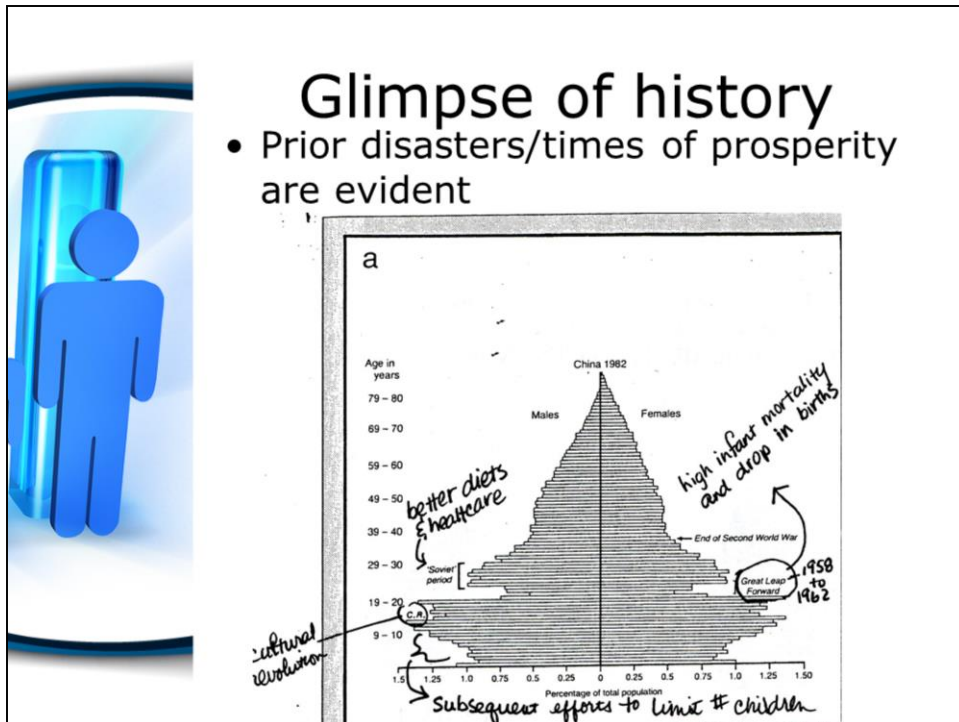
Population momentum is the time period that follows a country's TFR drop to replacement rate – there are still a large number of people in the population who continue to grow, so even if each if the early adults are only replacing themselves, the population continues to grow as they “age out” beyond the age of reproduction. It takes several decades after TFR reaches replacement rate for the population growth to level off.

Example – China instituted a “One Child” policy in the 1979. The TFR of China quickly fell to replacement rate over the next decade. However, even today, the population of China continues to grow as all the children who were born in the 1970s and early 1980s are just now reaching middle age, and their children have not yet reached the age to start their own families. China is still in the period of population momentum.

China's period of population momentum is actually short compared to those countries that go through the transition organically (without outside influence from government or other). Many parts of India have lowered their TRF to near replacement rates, but the country is expected to continue to grow until the population pyramid “fills out” – in other words, it completes the transition from steep-sided pyramid to box, which will take several generations.

“peak child” is another way to say replacement rate – what that means is that when an area hits “peak child” it has the maximum total number of children that there will ever be in this society. All generations before had fewer total children, and future generations will have fewer children as well because the adults are having smaller families. As a whole, the world hit “peak child” around the year 2000.

The phenomena of population momentum is why demographers can estimate that the world population will stop growing at around 11 billion people.



This population pyramid of China in 1982 shows several distinct periods in Chinese history.

The big gap in the cohort that was born in 1958-1962 shows a tremendous dip in the population – this is during what was known as the “Great Leap Forward” which was actually an attempt by the communist government of China with the goal of collectivization of agricultural and industrial output that led to a famine that left as many as 45 million dead. This dip in the population on the pyramid shows both a drop in the number of births, but also a spike in infant mortality rates during that time.

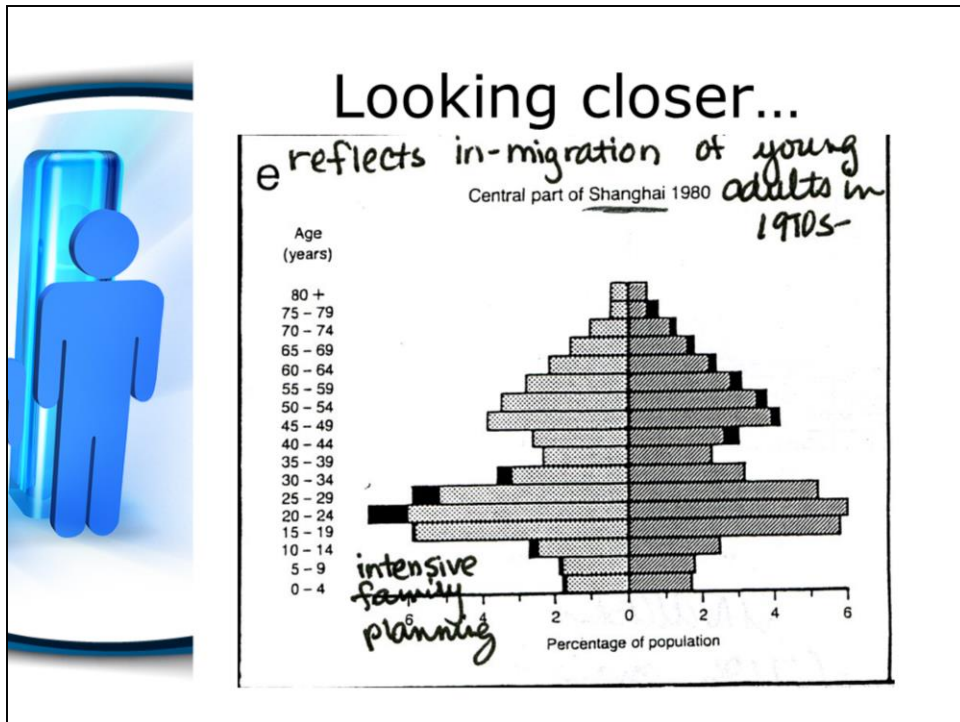
Following the Great Leap Forward, there was a baby boom (huge increase in births), where the pyramid bulges outward.

Then, in 1979, China instituted its One Child Policy to curb the rapid population growth by limiting families to have only one child (enforced mostly in cities through tax penalties and denying subsequent children access to services such as school and healthcare). As you can see on the graph, the age cohorts get smaller and smaller.



Analysis at different scales

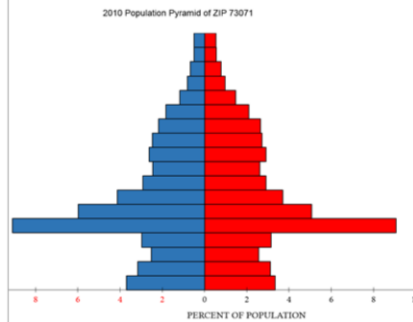
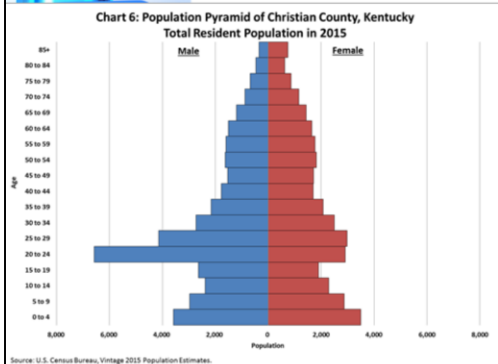
- Patterns of city/regional differ from larger areas
 - Unique needs of cities, communities
 - Help plan for future services in that place



For some, the pattern may not have a clear cause, but research on the history of the place may help you identify the pattern.

This is the Central part of Shanghai (a city in China) in 1980. Many of the young adult cohorts are recent migrants who left the rural areas of China to find work in the cities – there is a higher percentage of men (the black shading shows the difference) because young single men are more able to move to look for work than are married women or young single women who are expected to remain home and care for their parents until they marry.

Common City Pop. patterns

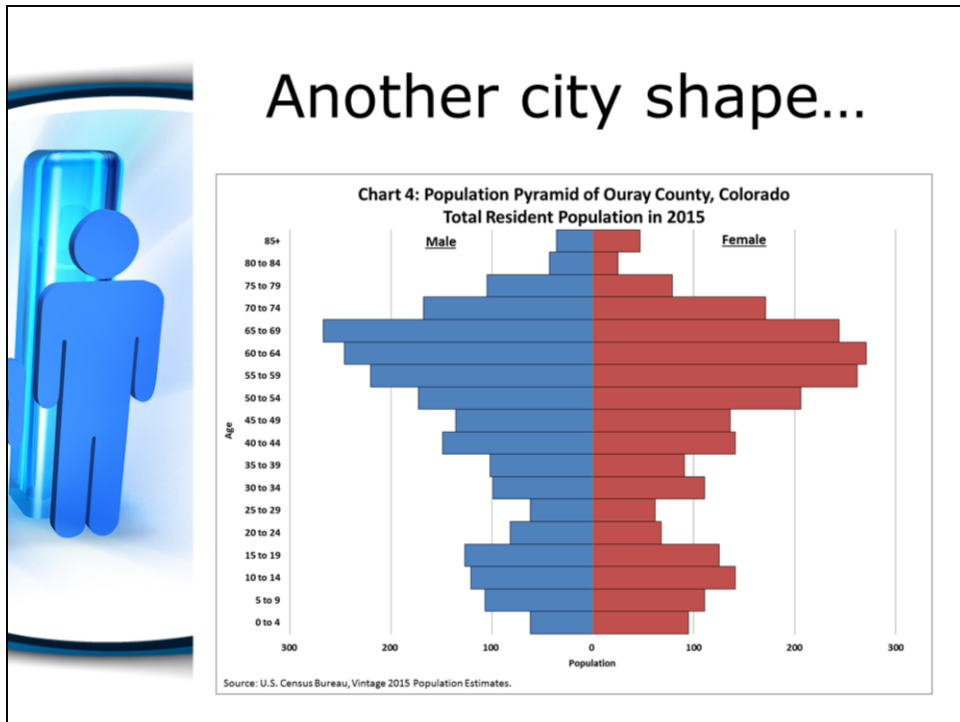


The two pyramids above both show a city.

The pyramid on the left has a disproportionate share of males aged 18-35, while the pyramid on the right has large numbers for both males and females within that age group. Why?

The pyramid on the left is a city with a military base.

The pyramid on the right is a city with a university.



Here's another city. What is a possible explanation for this weird shaped pyramid?

It is likely a retirement community, and the bulge around the young adult age are the workers who provide the services in the community.



Predict future issues

- Aging population?
 - Need for healthcare
 - Retirement systems/social security needs increase
 - People to fill jobs
- Youthful populations?
 - Need more jobs
 - Need more schools



WHERE will we need more schools?

WHERE will we need more jobs?

How about grocery stores, roads, hospitals, public transportation systems?

Looking at the population trends of a place begins with understanding the population composition of the area will help planners determine what services they will need in the future based on the age and trends within the population at that place.